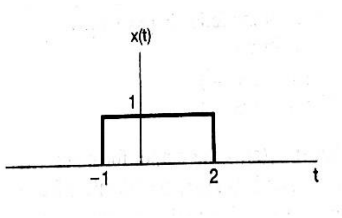
Signals and Systems Class Test 1

Name: **Roll No.:**

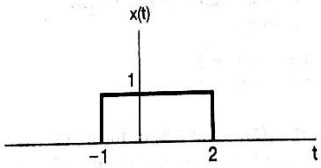
Q1. Consider a signal x(t)given in the figure below.



If $y(t) = x(t) * x(t)$, then value of the integral $\int_{-\infty}^{+\infty} y(t)dt$ is ...

(a)3 (b)9 (c)6 (d)12

Q2. A signal g(t)is defined as $g(t) = x(t) * u(t)$, where u(t) is the unit step signal and x(t)is shown in the figure below.



The signal g(t) is

(a) Periodic signal (b) an energy signal (c) neither energy nor power signal (d) a power signal

Q3. An LTI sustem given by

$$y(t) = \int_{t-\frac{T}{2}}^{t+\frac{T}{2}} x(\tau)d\tau \text{ is}$$

(a) Stable and Time varying (b) Unstable and Time varying

(c) Stable and Time invariant (d) Unstable and Time invariant.

Q4. The fundamental period of the signal $x(t) = 10 \cos^2(10 \pi t)$ is _____

a) 0.2 b) 0.1 c) 0.5 d) No fundamental period exists

Q5. Consider a discrete time signal $x[n] = 1 - \sum_{k=3}^{\infty} \delta[n-1-k]$. Find the values of M *and* n_0 so that $x[n] = u[Mn - n_0]$, where $u[n]$ is the discrete unit step signal.

a) $M = -1$, $n_0 = 3$ b) $M = -1$, $n_0 = -3$ c) $M = 1$, $n_0 = 3$ d) $M = 1$, $n_0 = -3$

Q6. Determine the power of the following signal

$y(t) = Ae^{jw0t}$

(a) ∞ (b) A (c) 0 (d) A^2

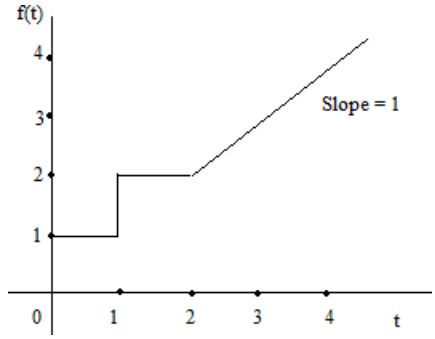
Q7. Which one of the following gives the correct description of the waveform shown in figure?

a) $u(t) + u(t-1)$

b) $u(t) + (t-1)u(t-1)$

c) $u(t) + u(t-1) + (t-2)u(t-2)$

d) $u(t) + (t-2)u(t-2)$



Q8. Determine whether the following signals are periodic or not?

$x1(t) = Even[cos(2\pi t)u(t)]$

$x2(n) = u(n) + u(-n)$

(a) aperiodic, aperiodic (b) periodic, periodic (c) periodic, aperiodic (d) aperiodic, periodic

Q9. A discrete time signal $x[n]$ is given by $x[n] = \{0, \underset{\uparrow}{0}, 1, 2, 3, 3, 0\}$. Which of the following is $x[2n]$?

a) $\{3, 2, \underset{\uparrow}{0}, 0\}$ b) $\{\underset{\uparrow}{0}, 2, 3, 0\}$ c) $\{3, 3, \underset{\uparrow}{1}, 0\}$ d) $\{\underset{\uparrow}{0}, 1, 3, 3\}$

Q10. Determine the energy of the signal $y(t)$ where,

$x(t) = \delta(t+1) - \delta(t-3)$ and $y(t) = \int_{-\infty}^{t} x(\alpha)d\alpha$

(a) 2 (b) 3 (c) 4 (d) 1

Q11 Consider a discrete time LTI system with impulse response $h[n] = (\frac{1}{3})^n u[n]$, The response of system to input $x[n] = (-1)^n$ is

a) $\frac{1}{3}(-1)^n$ b) $\frac{1}{4}(-1)^n$ c) $\frac{3}{4}(-1)^n$ d) $\frac{4}{3}(-1)^n$

Q12. The convolution, $x[n] * \delta[-n-1]$ is

a) $x[n-1]$ b) $x[n+1]$ c) $x[-n-1]$ d) $x[-n+1]$

Q13. The fundamental period of the signal $x[n] = 1 + \sin(\frac{\pi}{3}n) + \cos(\frac{\pi}{4}n)$ is:

a) 12 b) 24 c) 48 d) 36

Q14. Determine the output $y(n)$ of a LTI system with impulse response $h(n) = a^n u(n)$, $|a| < 1$ with the input sequence $x(n) = u(n)$.

a) $\frac{1-a^{n+1}}{1-a}$ b) $\frac{1-a^{n-1}}{1-a}$ c) $\frac{1+a^{n+1}}{1+a}$ d) $\frac{1+a^{n+1}}{1-a}$

Q15. Consider a system with input x[n] and output $y[n] = \sum_{k=-\infty}^{n} 2^{k-n} x(k+1)$. This system is:

a) Linear, Time Varying and Causal b) LTI and Causal c) Nonlinear and Time Varying d) LTI